

$t = -it_E$, with closed periodic paths

$$q(t_E + \beta) = q(\beta)$$

$$Z(\beta) = \text{tr}(e^{-\beta \hat{H}}) = \int Dq e^{-S_E[q]} \Big|_{q(t_E + \beta) = q(t_E)}$$

$$S[q] = \int d^4x \left[-\frac{1}{4} \partial_\mu q \partial^\mu q - \frac{1}{2} m^2 q^2 - V(q) \right]$$

* n-pt functions:

$$G_n(x_1, \dots, x_n) = \mathcal{Z}(\vec{T}) \langle \hat{\phi}(x_1) \dots \hat{\phi}(x_n) \rangle = \int D\phi e^{iS[\phi]} \phi(x_1) \dots \phi(x_n)$$

Note:

$$t_n \equiv x^0 = -x_0 = -it_E \quad \left. \vphantom{t_n} \right\} \quad x^4 = ix^0$$

$$t_E = x_4 = x^4$$

$$\text{and so } a_\mu b^\mu = a_\mu b_\nu \delta^{\mu\nu}$$

Euclidean Greens fⁿ:

$$G_n^{(E)}(x_1, \dots, x_n) = \int D\phi e^{-S_E[\phi]} \phi(x_1)$$

$$J \cdot \phi \equiv \int d^d x J(x) \phi(x)$$

$$G_n = \left(\frac{\delta}{\delta J(x_1)} \dots \frac{\delta}{\delta J(x_n)} \right) \int D\phi e^{-S_E + J \cdot \phi} \Big|_{J=0}$$

$$Z[J] = \int D\phi e^{-S_E} \sum_{n=0}^{\infty} \frac{1}{n!} (J \cdot \phi)^n$$

$$= \int D\phi e^{-S_E} \left(1 + J(x) \phi(x) + \dots \right)$$

$$= \sum_{n=0}^{\infty} J^n \int D\phi e^{-S_E} \phi^n$$

but

$$G^n(x_1, \dots, x_n) = \int D\phi e^{-S_E} \phi^n$$

and so we see the Generating fⁿ factors as

$$Z[J] = \sum_{n=0}^{\infty} \frac{1}{n!} \int \prod d^d x \cdot G(x_1, \dots, x_n) J(x_1) \dots J(x_n)$$

Perturbation Theory:

$$S[\phi] = \underbrace{S_0[\phi]}_{\text{free}} + \underbrace{S_I[\phi]}_{\text{interaction}}$$

Momentum Space Green's function

$$\tilde{G}_n(p_1, \dots, p_n) = \int d^d x_1 \dots d^d x_n e^{i p_1 x_1 + \dots} G_n(x_1, \dots, x_n)$$

* Note from translational invariance

$$x_i \rightarrow x_i - x_0$$

$$G(x_1, \dots, x_n) \rightarrow G(x_1 - x_0, x_2 - x_0, \dots)$$

If we choose

$$x_0 = x_1$$

$$G(x_1, \dots, x_n) = G(0, x_2 - x_1, \dots, x_n - x_1)$$

$$\tilde{G}_n(p_1, \dots, p_n) = \int d^d x_1 \dots d^d x_n e^{i(p_1 x_1 + \dots + p_n x_n)} G(0, x_2 - x_1, \dots)$$

Take integration variables

$$x_i \rightarrow x_i + x_1 \quad \text{for } i = 2, \dots, n$$

$$\begin{aligned} \tilde{G}_n(p_1, \dots, p_n) &= \int d^d x_1 \dots d^d x_n e^{i x_1 (p_1 + \dots + p_n)} e^{i (x_2 p_2 + \dots + p_n x_n)} \\ &\quad G(0, x_2, \dots, x_n) \\ &= (2\pi)^d \delta^d(p_1 + \dots + p_n) G_n(p_1, \dots, p_n) \end{aligned}$$

$|0\rangle \equiv$ Vacuum of free theory

$$\langle 0 | \mathcal{O}[\hat{\phi}] | 0 \rangle = \int \mathcal{D}\phi e^{-S_0[\phi]} \mathcal{O}[\phi]$$

• Now, consider

$$\mathcal{O}[\hat{\phi}] = e^{-S_I[\phi]}$$

Thus:

$$\langle 0 | e^{-S_F} | 0 \rangle = \int D\phi e^{-S}$$

and so the full Greens f^{μ} is given by the following:

$$G_n(x_1, \dots, x_n) = \langle 0 | \phi(x_1) \dots \hat{\phi}(x_n) e^{-S_F} | 0 \rangle \\ = \int D\phi e^S \phi_{x_1} \dots \phi_{x_n}$$

* The Generating f^{μ} is given by

$$Z[J] = \langle 0 | e^{-S_F} e^{\int d^d x J(x) \phi(x)} | 0 \rangle \\ = \int D\phi e^{-S_0 + J\phi} e^{-S_F}$$

Solution of the free field theory:

$$Z_0[J] = \int D\phi e^{-S_0 + J\phi} \\ = \int D\phi \exp \left\{ -\frac{1}{2} \int d^d x (\partial_\mu \phi \partial^\mu \phi + m^2 \phi^2) + \int d^d x J\phi \right\}$$

• Since F has no boundary terms

$$Z_0[J] = \int D\phi e^{-\frac{1}{2} \int d^d x (-\partial_\mu \partial^\mu + m^2) \phi + \int d^d x \phi(x) J(x)}$$

* Define: $(-\partial_\mu \partial^\mu + m^2) \equiv \Delta^{-1}$

$$e^{-\frac{1}{2} \int d^d x \phi \Delta^{-1} \phi + \int d^d x J(x) \phi(x)}$$

formally we write this as

$$\int D\phi e^{-\frac{1}{2} \phi \Delta^{-1} \phi + J\phi} = (2\pi)^{n/2} \det(\Delta^{-1})^{-1/2} e^{\frac{1}{2} J \Delta J} \\ N e^{\frac{1}{2} J \Delta J}$$

N is 'infinite' constant

* Complete the square: $(\phi - \alpha) a (\phi - \alpha) + \beta$

$$= -\frac{1}{2} \phi \Delta^{-1} \phi + J\phi$$

$$\therefore a = -\frac{1}{2} \Delta^{-1}$$

$$\beta = -a \alpha^2$$

$$-2\alpha a = J$$

$$\alpha = \Delta J$$

$$\Rightarrow -\frac{1}{2} (\frac{1}{2} \Delta^{-1}) (\Delta J)^2 = \beta$$

$$\alpha = \Delta J$$

$$\beta = \frac{1}{2} J \Delta J$$

$$\int D\phi e^{-\frac{1}{2} \phi \Delta^{-1} \phi + J\phi} = \int D\phi e^{-\frac{1}{2} (\phi - \Delta J) \Delta^{-1} (\phi - \Delta J) + \frac{1}{2} J \Delta J}$$

$$= e^{\frac{1}{2} J \Delta J} \int D\phi e^{-\frac{1}{2} (\phi - \Delta J) \Delta^{-1} (\phi - \Delta J)}$$

$$\phi' \equiv \phi - \Delta J$$

$$e^{\frac{1}{2} J \Delta J} \langle 0|0 \rangle_0 \equiv e^{\frac{1}{2} \int d^d x \int d^d y J(x) \Delta(x,y) J(y)} \langle 0|0 \rangle_0$$

$$Z[J] = \frac{Z_0[J]}{Z_0[0]} = e^{\frac{1}{2} J \Delta J}$$

We can solve $\Delta \Delta^{-1} = \delta(x-y)$

$$\Delta^{-1} = -\partial^\mu \partial_\mu + m^2$$

F.T: $(-\partial^\mu \partial_\mu + m^2) \tilde{\Delta}(p, y) = \int d^d x e^{-i p x} \delta(x-y)$

$$\tilde{\Delta}(p, y) = \frac{1}{p^2 + m^2} e^{-i p y} = e^{-i p y}$$

$$\Delta(x, y) = \int \frac{d^d p}{(2\pi)^d} \frac{e^{i p(x-y)}}{p^2 + m^2} \quad (\text{Euclidean})$$

* Going from $\mathbb{E}^d \rightarrow \mathbb{M}^d$:

leads to poles at $|p| = \pm i m$

* Need to rotate the po plane with $\pi/2 - \epsilon$

$$p^0 \rightarrow e^{i(\pi/2 - \epsilon)} p^0 = (i p^0) \bar{e}^{i\epsilon}$$

$$= i p_m^0 (1 - i\epsilon)$$

$$(p_E^0)^2 = -p_m^0{}^2 (1 - 2i\epsilon)$$

$$= -(p_m^0)^2 + 2i p_m^0 \epsilon$$

Because we are to regulate the higher energies we can assume

$$p_\mu \in \mathbb{R}^4$$

and so we choose an ϵ s.t.

$$\epsilon \equiv -2p_\mu^0 \in \mathbb{R}$$

and so

$$\Delta(t, \vec{x}) = \Delta_F(t, \vec{x})$$

Wick's Thm:

$$\text{ex: } f[\phi] = \phi_1 \phi_2^3 \phi_3^2$$

$$\langle f \rangle = \int \mathcal{D}\phi e^{-S_\phi} \phi_1 \phi_2^3 \phi_3^2 = \frac{\delta}{\delta J_1} \left(\frac{\delta}{\delta J_2} \right)^3 \left(\frac{\delta}{\delta J_3} \right)^2 \int \mathcal{D}\phi e^{-S_\phi}$$

$$\langle 0 | f[\phi] | 0 \rangle_J = f\left[\frac{\delta}{\delta J}\right] Z_0[J] \Big|_{J=0}$$

$$\text{Let } f[\phi] \equiv e^{-\int d^4x V(\phi)} \text{ then,}$$

$$Z[J] = e^{-\int d^4x V\left(\frac{\delta}{\delta J}\right)} Z_0[0] = e^{-\frac{1}{2} \int d^4x V\left(\frac{\delta}{\delta J}\right)} e^{\frac{1}{2} \int d^4x J \Delta J}$$

(Wick's thm in Path ML form)

$$\text{ex: } \lambda \phi^3/3! \quad , \quad V = \lambda \phi^3/3! \quad (\text{interaction})$$

$$Z[J] = \left(e^{-\int d^4x \lambda \left(\frac{\delta}{\delta J}\right)^3/3!} \right) Z_0$$

$$= \left(e^{-\int d^4x \lambda \left(\frac{\delta}{\delta J}\right)^3/3!} \right) Z_0$$

$$= e^{-\int d^4x \lambda \left(\frac{\delta}{\delta J}\right)^3/3!} e^{\frac{1}{2} \int d^4x J \Delta J}$$

$$= \left(1 - \lambda \left(\frac{\delta}{\delta J}\right)^3 + \lambda^2 \left(\frac{\delta}{\delta J}\right)^3 \left(\frac{\delta}{\delta J}\right)^3 + \lambda^3 \left(\frac{\delta}{\delta J}\right)^3 \dots \right) Z_0[J]$$

acting on $Z_0[J]$

$$\begin{aligned}
 \textcircled{a}: \left(\frac{\delta}{\delta J_x} \right)^3 Z[J] \Big|_{J=0} &= \left(\frac{\delta}{\delta J_x} \right)^2 \left((\Delta_{xy} J^y) e^{\frac{i}{2} J \Delta J} \right) \\
 &= \frac{\delta}{\delta J_x} \left(\Delta_{xx} + \Delta_{xy} J^y \Delta_{xz} J^z \right) e^{\frac{i}{2} J \Delta J} \\
 &= \{ 2 \Delta_{xx} \Delta_{xz} J^z + (\Delta_{xz} J^z)^2 \} e^{\frac{i}{2} J \Delta J}
 \end{aligned}$$

... other stuff... i'll do this on paper.